CS3423 - Compilers II

Mini-Assignment 5

Kaleidoscope

DEADLINE: 11:59 PM, 27th October, 2026 (Tuesday)

Instructions:
Clone this repository for the assignment: sreevyshnavi02/kaleidoscope_tutorial
Make changes to this code for both the questions.

Questions:

1. Consider adding two new binary operators to Kaleidoscope DSL - division (/) and greater than (>). Make all necessary changes across the lexer, parser and code generation.

2. As discussed in the class, Kaleidoscope currently only supports variables of type double. This lets us write code in the DSL without mentioning the type anywhere, and the lowering to LLVM IR is also done assuming that the type is always a double. This question requires you to extend Kaleidoscope DSL to support two more types: int, bool.
   Make necessary changes to allow the user to specify the types for every variable definition, and record the type of the variable in the symbol table (recall from the class that NamedValues in toy.cpp is the symbol table currently used, extend it).

   Add the required semantic checks for those data types such as -
   a. Type checking
   b. Type casting

Submission Details:
Teams of 2 people are allowed. Your submission should be a gzip archive with the name: Mini-Asgn-5-ROLLNO1-ROLLNO2.tar.gz

The unzipped directory should contain:
Report.pdf - briefly explaining the changes you made
Kaleidoscope directory (cloned from github with the required modifications)